

CASE STUDY 117

VELLUM ANTIQUARIAN BOOKSTORE & RARE BOOKS CHAIN

Campus LAN Segmentation, Trunking & Spanning Tree

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1. INTRODUCTION

Vellum Antiquarian Bookstore operates three locations: Main Street, Harborfront and Auction Annex. The three locations are interconnected in a triangular topology. The existing flat network creates excessive broadcast traffic and allows guest devices to potentially access internal resources.

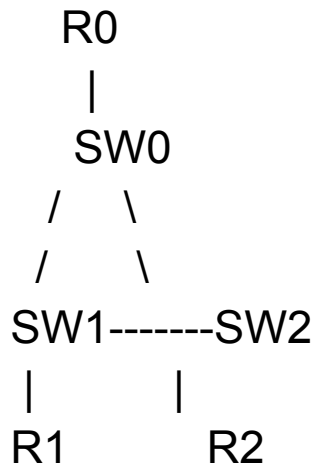
The proposed network redesign uses VLAN segmentation, 802.1Q trunking, Spanning Tree Protocol, EtherChannel, router-on-a-stick inter-VLAN routing and Layer-2 security controls.

2. NETWORK TOPOLOGY

The Packet Tracer topology consists of three routers and three switches:

- R0 + SW0 – Main Street
- R1 + SW1 – Harborfront
- R2 + SW2 – Auction Annex

The switches are connected in a triangle:



Two parallel physical links are provided between SW0 and SW2 for the proposed EtherChannel.

A total of 12 PCs are connected to their respective local switches.

3. VLAN DESIGN

VLAN 13 – Provenance-Systems

Used at Main Street and Auction Annex for cataloguing terminals and authentication systems.

VLAN 23 – Staff-POS

Used at all three locations for POS terminals and staff devices.

VLAN 33 – Admin

Used at Harborfront for finance and administrative systems.

VLAN 43 – IT-Mgmt

Reserved across all locations for switch and network-management devices.

VLAN 915 – Guest

Used at Harborfront for public Wi-Fi devices.

VLAN 999 – Native

Unused VLAN used as the native VLAN on trunk links.

4. VLAN PORT ASSIGNMENT

SW0 – Main Street

Fa0/4–Fa0/5 → VLAN 13

Fa0/6–Fa0/7 → VLAN 23

SW1 – Harborfront

Fa0/4–Fa0/5 → VLAN 23

Fa0/6–Fa0/7 → VLAN 915

SW2 – Auction Annex

Fa0/4–Fa0/5 → VLAN 13

Fa0/6–Fa0/7 → VLAN 23

The remaining VLANs are maintained for the complete network design and future management devices.

5. TRUNKING

The inter-switch connections are configured as 802.1Q trunks. VLAN 999 is selected as the native VLAN and is deliberately unused for normal user traffic.

Using a dedicated unused native VLAN reduces the possibility of normal user traffic being associated with the native VLAN and provides an additional Layer-2 security measure against VLAN-hopping techniques involving native VLAN traffic. The native VLAN should remain unused for end-user devices.

The trunks carry VLANs 13, 23, 33, 43, 915 and 999.

6. ETHERCHANNEL DESIGN

Two parallel links between SW0 and SW2 are intended to be bundled into EtherChannel using LACP.

Physical links:

SW0 Fa0/2 ↔ SW2 Fa0/2

SW0 Fa0/3 ↔ SW2 Fa0/3

The resulting logical interface is Port-Channel 1.

EtherChannel allows the two physical links to operate as one logical connection. STP therefore treats the EtherChannel as a single logical path instead of independently blocking one of the physical links. This provides additional bandwidth while retaining link redundancy.

Implementation note: the EtherChannel configuration was started in Packet Tracer but was not successfully verified as operational before completion of the lab session.

7. SPANNING TREE DESIGN

The three switches create a Layer-2 triangle, which introduces a potential switching loop. Rapid PVST can be used to maintain a loop-free topology.

SW0 is designated as the intended root bridge.

Suggested bridge priorities:

SW0 – 4096

SW1 – 8192

SW2 – 12288

With SW0 as the root bridge, STP selects the shortest paths toward SW0 and places a redundant path into a blocking/discarding state where required.

The EtherChannel between SW0 and SW2 is treated as one logical STP connection after successful bundling.

8. INTER-VLAN ROUTING

Router-on-a-stick is proposed for communication between VLANs.

Each router uses a trunk connection to its local switch and subinterfaces for the required VLANs.

Example gateway structure:

VLAN 13 → 192.168.13.1/24

VLAN 23 → 192.168.23.1/24

VLAN 33 → 192.168.33.1/24

VLAN 43 → 192.168.43.1/24

VLAN 915 → 192.168.91.1/24

The appropriate default gateway is assigned to each end device according to its VLAN.

9. GUEST NETWORK SECURITY

Harborfront Guest devices are placed in VLAN 915. An ACL is proposed to prevent Guest traffic from reaching the Provenance-Systems VLAN 13.

The security policy is:

Guest VLAN → Provenance VLAN = DENY

Guest VLAN → Other permitted networks = ALLOW

This prevents public Wi-Fi users from accessing sensitive inventory and provenance systems while allowing normal permitted network communication.

10. PORT SECURITY

Port security is proposed on Harborfront Guest access ports.

Configuration requirements:

Maximum MAC addresses: 2

Violation mode: restrict

MAC learning: sticky

The restrict mode is appropriate for a public environment because a violation can be recorded and offending traffic can be discarded without immediately shutting down the entire

switch port. This reduces disruption when customers change devices while still enforcing the configured MAC-address limit.

11. VTP RECOMMENDATION

VTP Transparent mode is recommended for this small three-site environment.

In transparent mode, VLAN information can be maintained locally on each switch rather than relying on a centralized VTP server. This reduces the risk of an incorrect VLAN database change being propagated throughout the network.

The trade-off is that VLAN configuration must be maintained manually on the switches. For only three switches and a relatively small VLAN scheme, this additional administration is manageable.

12. LINK-FAILURE SCENARIO

The network contains redundant paths between the three switches.

If the SW0–SW1 connection fails, STP detects the loss of the active path. The previously redundant path through SW2 can then transition to the required forwarding state after STP reconvergence.

The resulting path becomes:

SW1 → SW2 → SW0

This allows communication to continue without maintaining a permanent Layer-2 loop.

The exact port state transition should be verified in Packet Tracer using:

`show spanning-tree`

and by observing the interface state before and after shutting down the SW0–SW1 link.

13. VERIFICATION COMMANDS

The following commands are used to verify the completed design:

`show vlan brief`

`show interfaces trunk`

`show etherchannel summary`

`show spanning-tree`

`show ip interface brief`

`show running-config`

`show port-security`

`show port-security interface`

Connectivity can be tested using ping and traceroute between appropriate VLANs.

The Guest VLAN should specifically be tested against the Provenance VLAN to confirm that the ACL blocks the prohibited traffic.

14. IMPLEMENTATION STATUS

The physical Packet Tracer topology was created with:

- Three routers
- Three switches
- Three interconnected sites
- Twelve PCs
- Triangle switch topology
- Two parallel SW0–SW2 links

The VLANs were created and PC access ports were assigned. Inter-switch trunk configuration was also performed.

The EtherChannel configuration was attempted using LACP but was not successfully verified as operational before the practical session was stopped. The remaining STP, inter-VLAN routing, ACL, port-security, VTP and failure-testing stages remain part of the proposed design and configuration plan.

15. CONCLUSION

The proposed Vellum Antiquarian Bookstore network replaces the original flat LAN with a segmented and redundant architecture. VLANs separate business functions, trunks transport VLAN traffic between locations, STP prevents Layer-2 loops, EtherChannel provides redundant bandwidth, router-on-a-stick enables controlled inter-VLAN communication, and security controls protect the Guest and Provenance networks.

The design provides a structured foundation for a scalable three-site bookstore network while addressing broadcast traffic, segmentation, redundancy and security requirements.